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1) Bayes rule: $P(h|D) = \frac{P(D|h) \cdot P(h)}{P(D)}$

a/ $P(h|D)$: Posterior; $P(h)$: Prior; $P(D)$: Evidence

$P(D|h)$: Likelihood.

b) $h_MAP = \text{argmax}\ P(D|h) \cdot P(h)$

$h_ML = \text{argmax}\ P(D|h)$

They are equal when prior is the same for all hypothesis.

c/ When training set is small, ~~the Prior~~ h_MAP is more depend on the Prior so the Posterior may differ compared with h_ML

2/a/ $P(\text{disease}) = 0,01$; $P(+|\text{disease}) = 0,95$; $P(+|\text{no disease}) = 0,05$

$P(+) = P(+|\text{disease}) \cdot P(\text{disease}) + P(+|\text{no disease}) \cdot P(\text{no disease})$

$= 0,95 \cdot 0,01 + 0,05 \cdot 0,99 = 0,059$

$P(\text{disease}|+) = \frac{P(+|\text{disease}) \cdot P(\text{disease})}{P(+)} = \frac{0,95 \cdot 0,01}{0,059} \approx 0,161$

b/ h_{ML} compares the Likelihoods: $P(+|\text{disease})$ vs $P(+|\text{no disease})$

$\rightarrow$ It picks disease — $0,95$ — $0,05$

h_{MAP} compares the Posteriors: $P(\text{disease}|+)$ vs $P(\text{no disease}|+)$

$\rightarrow$ It picks no disease. — $0,161$ — $0,839$

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3/a/ Likelihood: $0,02; 0,1; 0,3 \rightarrow h_{ML} = 0,3\ (h_3)$

b/

h	$P(h) \cdot P(D\|h)$	Posterior
h_1	$0,3 \cdot 0,02 = 0,014$	$0,014/0,064 = 0,219$
h_2	$0,2 \cdot 0,1 = 0,02$	$0,02/0,064 = 0,313$
h_3	$0,1 \cdot 0,3 = 0,03$	$0,03/0,064 = 0,469$

$h_{MAP} = h_3$

c/ They are the same in this case (h_3), maybe because the prior is low.

4/a/ $\hat{y} = \text{argmax}_y\ P(y) \prod_i P(x_i|y)$

b/ Counts: Yes = 5, No = 3

	Yes	No
P(Play)	5/8	3/8
Outlook = Sunny	2/5	1/3
Outlook = Overcast	1/5	0/3
Outlook = Rain	2/5	2/3
Wind = Weak	4/5	0/3
Wind = Strong	1/5	3/3

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No.

Date

c) For (Overcast, Strong)

. Yes : 5/8 . 1/5 . 1/5 = 0,025

. No : 3/8 . 0 . 1 = 0

→ Prediction is Yes.

d/ Because Overcast never ~~appea~~ appears with No in the Data.

e/ Not sure

5/a. Naive Bayes : 1 prior parameter.

. Full joint : $2^{n+1} - 1$ parameters.

b/ Not sure.

6/ The product of many small probabilities become very small so in order for computer to keep track of that underflow issue, we turn that product to sum of logs, the argmax is unchanged.

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